IMPORTANT DIFFERENCES between LOG files from [mcr_2405] –vs– [2603] versions of DMON firmware/d3 programs:

- **[2603]** version has **49 characters** between "$DUTY" and "*" in each line that get decoded *([mcr_2405] has 41 characters)*
- **[2603]** version uses **MICROseconds** represented by **8-character** field *([mcr_2405] uses millisec 4-char field; **1 milisec = 1000 microsec**)*

UVI2_PRSTT_North_20260428 mission used **[2603]** version DMON update release
*(firmware **mcrec2603.bin**, d3host.exe **version 2.4**, d3read.exe **version 2.1**, and reformat_dmon_wav_files.sav **version 2**)*

Example $DUTY line in log file (from UVI2_PRSTT_North_20260428_050.log):

1778722792,53,$DUTY,6A0527A9000F3A836A0527E5000F3A7F30000104A00DBBA00*00

*The **49** characters between "$DUTY" and "*" in each line can be decoded as follows…*

Characters 1-8 (n=8): Start datetime in Unix epoch UTC

6A0527A9 hexadecimal = **1778722729** decimal (sec Unix epoch) = **2026-05-14 01:38:49** start datetime UTC

Characters 9-16 (n=8): Microseconds after start dt in Unix epoch UTC

000F3A83 hex = **998019** decimal (microsec) →

2026-05-14 01:38:49 UTC + 998019 microsec = **2026-05-14 01:38:49.998019** precise start dt UTC

Characters 17-24 (n=8): End datetime in Unix epoch UTC

6A0527E5 hex = **1778722789** decimal (sec Unix epoch) = **2026-05-14 01:39:49** end datetime UTC

Characters 25-32 (n=8): Microseconds after end dt in Unix epoch UTC

000F3A7F hex = **998015** decimal (microsec) →

2026-05-14 01:39:49 UTC + 998015 microsec = **2026-05-14 01:39:49.998015** precise end dt UTC

Character 33 (n=1): Hydrophone numeric ID **2 = LF hydrophone, 3 = HF hydrophone, 1 = MF hydrophone*

3 numeric ID (does NOT require decoding) = HF hydrophone

Characters 34-41 (n=8): Sequential recording session number for **specific hydrophone (LF/HF)* during mission

0000104A hex = **4170** decimal = **4,170**th recording session for HF hydrophone during mission

Characters 42-49 (n=8): Number of samples in current recording session

00DBBA00 hex = **14400000** decimal = **14,400,000** total samples in the 4,170th HF recording session

OPTIONAL: Check if n samples matches expected value calculated from duration and sample rate

- Duration = 2026-05-14 01:38:49.998019 UTC to 2026-05-14 01:39:49.998015 UTC = 59.999996 sec
- Sample rate = 240 kHz = 240,000 Hz = 240,000 samples/sec
- Expected total samples = 59.999996 sec × 240,000 samples/sec = 14,399,999.04 samples
  OFF by 0.96000000089 samples